

Tension Prediction in Non-Tonal Music: A Comparative Study

(1) Abstract

Tension is a highly complex aspect of music perception. The questions of defining tension and identification of musical parameters and structures that contribute to perceived tension have been an interdisciplinary area feeding from music theory, cognitive psychology and acoustics. Quantitative methods of calculating expected perceived tension based on musical parameters culminated in the advent of tension models, such as Lerdahl's [1] and Farbood's [2]. With the recent advancements in music information retrieval, we see that tension prediction algorithms such as Barchet et al.'s TenseMusic [3] are able to implement these models through raw audio analysis. These models have shown demonstrable success in predicting tension in tonal music which they are conceived to do, however their applicability in non-tonal or Non-Western musical contexts leaves questions. The fundamental tension prediction models of Lerdahl and others are heavily dependent on changes in harmony and melody as principal parameters. TenseMusic study shows disagreement with the results of its behavioral ratings and the algorithm in less conventional pieces.

The purpose of this project is to identify and quantify aspects of non-tonal music that cause the state of the art tension prediction algorithms to deliver suboptimal results. For this, we utilize TenseMusic to analyse a selection of repertoire of non-tonal music from the 20th century. Afterwards, we compare the tension curves created by the algorithm to the results of the real-time tension perception ratings we gathered from 30 participants. Then, we identify the musical parameters that could have contributed to agreement or disagreement between the experiment and the algorithm, arguing for potential modifications in the given models for non-tonal optimization.

(2) Experiment Description

The experiment consists of participants rating their perceived tension continuously while the musical repertoire is presented to them. Data will be collected using PsychoPy¹ running on a Pavlovia server, which allows the real-time rating the experiment demands. This will be facilitated online remotely. Tension will be assessed using continuous ratings, a validated method from previous studies [3]. Participants will use headphones and at the first step, will adjust their volume while listening to a jazz excerpt. Afterwards, the musical pieces will be presented in randomized order together with the digital slider. Participants will move the digital slider one way or another while they listen to the preplanned selection of music. Demographic information including age and musical experience will be collected prior to starting the tension rating task. The preplanned selection of music consists of shorter than a minute excerpts from 30 compositions, making the experiment a roughly 45 minute process in total.

For recruitment, contact and correspondence for the the study, the researcher's University of Sussex e-mail address (a.kuru@sussex.ac.uk) will be used.

[1] Lerdahl F, Krumhansl CL. Modeling Tonal Tension. *Music Perception*. 2007; 24(4):329–366.
<https://doi.org/10.1525/mp.2007.24.4.329>

¹ <https://psychopy.org/about/index.html>

[2] Farbood MM. A Parametric, Temporal Model of Musical Tension. *Music Perception*. 2012; 29(4):387–428. <https://doi.org/10.1525/mp.2012.29.4.387>

[3] Barchet AV, Rimmele JM, Pelofi C (2024) TenseMusic: An automatic prediction model for musical tension. *PLoS ONE* 19(1): e0296385. <https://doi.org/10.1371/journal.pone.0296385>